

P0593R5

Implicit creation of objects for low-level object manipulation

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Abstract

This paper proposes that objects of sufficiently trivial types be created on-demand as necessary within newly-allocated storage to give programs defined behavior.

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This paper resolves CWG2325.

§ 1. Change history

Since [\[P0593R0\]](#):

- Paper expanded from Ville's original call for solutions to a description of a proposed solution, based on SG12 discussion.

Since [\[P0593R1\]](#):

Incorporated further SG12 feedback:

- An explicit syntactic marker is required to indicate that objects should be created. Existing obvious markers, such as the use of `malloc`, or simply performing member access on a union, suffice.
- Expand set of implicit-lifetime types to require *either* a trivial default constructor *or* a trivial copy/move constructor, rather than requiring both.
 - Types with only a trivial default constructor may be suitable for member-by-member construction via class member access, even if the copy or move constructor is non-trivial.
 - Types with only a trivial copy/move constructor may be suitable for initialization by copying (for example) an on-disk representation into memory, even if the default constructor is non-trivial.
- Define the C standard library `memcpy` and `memmove` functions as triggering implicit object creation.
- Add description of suggested "typed" form of `std::bless`.

Since [\[P0593R2\]](#):

- Removed union member access being sufficient to implicitly create objects based on objections from an implementer.

Since [\[P0593R3\]](#):

- Incorporated EWG feedback:
 - Aggregates are considered implicit-lifetime types
 - Provide a typed version of `std::bless`
- Added wording.

Since P0594R4:

- Incorporated LEWG feedback:

- Removed untyped version of `std::bless(p, n)`, favoring use of `new (p) std::byte[n]` in its place.
- Renamed typed version of `std::bless<T>` to `std::start_lifetime_as<T>`.
- Add `volatile` variant of typed version.
- Added `std::bit_cast` as an operation that implicitly creates objects.
- Added section on trivial union copies and added wording to copy the subobject structure.
- Separated standard library extensions from DR-level language rules fix.

§ 2. Motivating examples

§ 2.1. Idiomatic C code as C++

Consider the following natural C program:

```
struct X { int a, b; };
X *make_x() {
    X *p = (X*)malloc(sizeof(struct X));
    p->a = 1;
    p->b = 2;
    return p;
}
```

When compiled with a C++ compiler, this code has undefined behavior, because `p->a` attempts to write to an `int` subobject of an `X` object, and this program never created either an `X` object nor an `int` subobject.

Per [intro.object]p1,

An *object* is created by a definition, by a *new-expression*, when implicitly changing the active member of a union, or when a temporary object is created.

... and this program did none of these things.

§ 2.2. Objects provided as byte representation

Suppose a C++ program is given a sequence of bytes (perhaps from disk or from a network), and it knows those bytes are a valid representation of type `T`. How can it efficiently obtain a `T*` that can be legitimately used to access the object?

Example: (many details omitted for brevity)

```
void process(Stream *stream) {
    unique_ptr<char[]> buffer = stream->read();
    if (buffer[0] == F00)
        process_foo(reinterpret_cast<Foo*>(buffer.get())); // #1
    else
        process_bar(reinterpret_cast<Bar*>(buffer.get())); // #2
}
```

This code leads to undefined behavior today: within `Stream::read`, no `Foo` or `Bar` object is created, and so any attempt to access a `Foo` object through the `Foo*` produced by the cast at #1 would result in undefined behavior.

§ 2.3. Dynamic construction of arrays

Consider this program that attempts to implement a type like `std::vector` (with many details omitted for brevity):

```
template<typename T> struct Vec {
    char *buf = nullptr, *buf_end_size = nullptr, *buf_end_capacity = nullptr;
    void reserve(std::size_t n) {
        char *newbuf = (char*)::operator new(n * sizeof(T), std::align_val_t(alignof(T)));
        std::uninitialized_copy(begin(), end(), (T*)newbuf); // #a

        ::operator delete(buf, std::align_val_t(alignof(T)));
        buf_end_size = newbuf + sizeof(T) * size(); // #b
        buf_end_capacity = newbuf + sizeof(T) * n; // #c
        buf = newbuf;
    }
    void push_back(T t) {
        if (buf_end_size == buf_end_capacity)
            reserve(std::max<std::size_t>(size() * 2, 1));
        new (buf_end_size) T(t);
        buf_end_size += sizeof(T); // #d
    }
    T *begin() { return (T*)buf; }
    T *end() { return (T*)buf_end_size; }
    std::size_t size() { return end() - begin(); } // #e
};

int main() {
    Vec<int> v;
    v.push_back(1);
    v.push_back(2);
    v.push_back(3);
    for (int n : v) { /*...*/ } // #f
}
```

In practice, this code works across a range of existing implementations, but according to the C++ object model, undefined behavior occurs at points #a, #b, #c, #d, and #e, because they attempt to perform pointer arithmetic on a region of allocated storage that does not contain an array object.

At locations #b, #c, and #d, the arithmetic is performed on a `char*`, and at locations #a, #e, and #f, the arithmetic is performed on a `T*`. Ideally, a solution to this problem would imbue both calculations with defined behavior.

§ 3. Approach

The above snippets have a common theme: they attempt to use objects that they never created. Indeed, there is a family of types for which programmers assume they do not need to explicitly create objects. We propose to identify these types, and carefully carve out rules that remove the need to explicitly create such objects, by instead creating them implicitly.

§ 3.1. Affected types

If we are going to create objects automatically, we need a bare minimum of the following two properties for the type:

- 1) *Creating an instance of the type runs no code.* For class types, having a trivially default constructible type is often the right constraint. However, we should also consider cases where initially creating an object is non-trivial, but copying it (for instance, from an on-disk representation) is trivial.
- 2) *Destroying an instance of the type runs no code.* If the type maintains invariants, we should not be implicitly creating objects of that type.

Note that we're only interested in properties of the object itself here, not of its subobjects. In particular, the above two properties always hold for array types. While creating or destroying array elements might run code, creating the array object (without its elements) does not. For similar reasons, it also seems

reasonable to permit implicit object creation for aggregate class types even if the aggregate contains an element with a non-trivial destructor.

This suggests that the largest set of types we could apply this to is:

- Scalar types
- Aggregate types (arrays with any element type, aggregate classes with any members)
- Class types with a trivial destructor and a trivial constructor (of any kind)

(Put another way, we can apply this to all types other than function type, reference type, `void`, and class types where all constructors are non-trivial or where the destructor is non-trivial.)

We will call types that satisfy the above constraints *implicit-lifetime types*.

§ 3.2. When to create objects

In the above cases, it would be sufficient for `malloc` / `::operator new` to implicitly create sufficient objects to make the examples work. Imagine that `malloc` could "look into the future" and see how its storage would be used, and create the set of objects that the program would eventually need. If we somehow specified that `malloc` did this, the behavior of many C-style use cases would be defined.

On typical implementations, we can argue that this is not only natural, it is in some sense the status quo. Because the compiler typically does not make assumptions about what objects are created within the implementation of `malloc`, and because object creation itself typically has no effect on the physical machine, the compiler must generate code that would be correct if `malloc` did create that correct set of objects.

However, this is not always sufficient. An allocation from `malloc` may be sequentially used to store multiple different types, for instance by way of a memory pool that recycles the same allocation for multiple objects of the same size. It should be possible to grant such cases the same power to implicitly create objects as is de facto granted to `malloc`.

We could specify that implicit object creation happens automatically at any program point that relies on an object existing. This has a great deal of appeal: no explicit program action is ever required to create objects, and it directly describes a simple model where objects are not distinguished from the storage they occupy (this model gives the same results as C's "effective type" model in most cases). However, it also removes much of the power of scalar type-based alias analysis. The C committee has long been struggling with the conflict between their desire to support TBAA and their version of this rule, as exemplified by C's DR 236 ([C236]), which lists a "resolution" not reflected by the standard wording and that undesirably grants special powers to function call boundaries (this is one of at least four different and incompatible rules the C committee has at one point or another taken as the resolution to that defect). The lack of a reasonable resolution to these problems, despite them being known for nearly two decades, suggests that this is not a good path forward.

Therefore we propose the following rule:

Some operations are described as implicitly creating objects within a specified region of storage. The abstract machine creates objects of implicit-lifetime types within those regions of storage as needed to give the program defined behavior. For each operation that is specified as implicitly creating objects, that operation implicitly creates zero or more objects in its specified region of storage if doing so would give the program defined behavior. If no such sets of objects would give the program defined behavior, the behavior of the program is undefined.

The coherence of the above rule hinges on a key observation: changing the set of objects that are implicitly created can only change whether a particular program execution has defined behavior, not what the behavior is.

We propose that at minimum the following operations be specified as implicitly creating objects:

- Creation of an array of `char`, `unsigned char`, or `std::byte` implicitly creates objects within that array.

- A call to `malloc`, `calloc`, `realloc`, or any function named `operator new` or `operator new[]` implicitly creates objects in its returned storage.
- `std::allocator<T>::allocate` likewise implicitly creates objects in its returned storage; the allocator requirements should require other allocator implementations to do the same.
- A call to `memmove` behaves as if it
 1. copies the source storage to a temporary area
 2. implicitly creates objects in the destination storage, and then
 3. copies the temporary storage to the destination storage.

This permits `memmove` to preserve the types of trivially-copyable objects, or to be used to reinterpret a byte representation of one object as that of another object.

- A call to `memcpy` behaves the same as a call to `memmove` except that it introduces an overlap restriction between the source and destination.
- A call to `std::bit_cast` implicitly creates objects in the result, to handle the case where the destination type contains a union.
- A new barrier operation (distinct from `std::launder`, which does not create objects) could be introduced to the standard library, with semantics equivalent to a `memmove` with the same source and destination storage. Prior versions of this document suggested:

```
// Requires: [start, (char*)start + length) denotes a region of allocated
// storage that is a subset of the region of storage reachable through start.
// Effects: implicitly creates objects within the denoted region.
void std::bless(void *start, size_t length);
```

However, LEWG review observed that `new (start) std::byte[length]` can be used to obtain the desired effect by reusing the first rule above.

In addition to the above, an implementation-defined set of non-standard memory allocation and mapping functions, such as `mmap` on POSIX systems and `VirtualAlloc` on Windows systems, should be specified as implicitly creating objects.

Note that a pointer `reinterpret_cast` is not considered sufficient to trigger implicit object creation.

§ 3.3. Type punning

We do not wish examples such as the following to become valid:

```
float do_bad_things(int n) {
    alignof(int) alignof(float)
    char buffer[max(sizeof(int), sizeof(float))];
    *(int*)buffer = n;          // #1
    new (buffer) std::byte[sizeof(buffer)];
    return (*float*)buffer;    // #2
}
```

```
float do_bad_things(int n) {
    union { int n; float f; } u;
    u.n = n;          // #1
    new (&u) std::byte[sizeof(u)];
    return u.f;      // #2
}
```

The proposed rule would permit an `int` object to spring into existence to make line #1 valid (in each case), and would permit a `float` object to likewise spring into existence to make line #2 valid.

However, these examples still do not have defined behavior under the proposed rule. The reason is a consequence of [basic.life]p4:

The properties ascribed to objects and references throughout this document apply for a given object or reference only during its lifetime.

Specifically, the value held by an object is only stable throughout its lifetime. When the lifetime of the `int` object in line #1 ends (when its storage is reused by the `float` object in line #2), its value is gone. Symmetrically, when the `float` object is created, the object has an indeterminate value ([dcl.init]p12), and therefore any attempt to load its value results in undefined behavior.

Thus we retain the property (essential to modern scalar type-based alias analysis) that loads of some scalar type can be considered to not alias earlier stores of unrelated scalar types.

§ 3.4. Union copies

Consider an example such as:

```
union U {
    int n;
    float f;
};
float pun(int n) {
    U u = {.n = n};
    U u2 = u;    // #1
    return u2.f; // #2
}
```

In the current language rules, a strict interpretation of the wording would suggest that only the object representation of `u` is copied on line #1, but no union member's lifetime begins, so `u2` has no active union member. This is clearly not the appropriate outcome.

We could rectify this in one of two natural ways:

- Line #1 copies the object structure of `u` to `u2`, so that the active member of `u2` is `n`; line #2 does not have defined behavior, just as if it returned `u.f`.
- Line #1 implicitly creates objects; at line #2 we have implicitly bit-cast `n` to `float`.

This paper proposes we adopt the former option, as it preserves equational reasoning and results in more explicit code (that is, using `std::bit_cast` to perform bit-casts rather than union copies).

§ 3.5. Constant expressions

Constant expression evaluation is currently very conservative with regard to object creation. There is a tension here: on the one hand, constant expression evaluation gives us an opportunity to disallow runtime program semantics that we consider undesirable or problematic, and on the other hand, users strongly desire a full compile-time evaluation mechanism with the same semantics as the base language.

Following the existing conservatism in constant expression evaluation, and specific concerns about high implementation costs in some existing implementations, we propose that implicit creation of objects should *not* be performed during such evaluation. The disallowance of pointer or reference `reinterpret_cast`s in constant expressions is believed to make the lack of implicit object creation unobservable.

§ 3.6. Pseudo-destructor calls

In the current C++ language rules, "pseudo-destructor" calls may be used in generic code to allow such code to be ambivalent as to whether an object is of class type:

```
template<typename T> void destroy(T *p) { p->~T(); }
```

When `T` is, say, `int`, the pseudo-destructor expression `p->~T()` is specified as having no effect. We believe this is an error: such an expression should have a lifetime effect, ending the lifetime of the `int` object. Likewise, calling a destructor of a class object should always end the lifetime of that object, regardless of whether the destructor is trivial.

This change improves the ability of static and dynamic analysis tools to reason about the lifetimes of C++ objects.

§ 3.7. Practical examples

```
std::vector<int> vi;
vi.reserve(4);
vi.push_back(1);
int *p = &vi.back();
vi.push_back(2);
vi.push_back(3);
int n = *p;
```

Within the implementation of `vector`, some storage is allocated to hold an array of up to 4 `ints`. Ignoring minor differences, there are two ways to create implicit objects to give the execution of this program defined behavior: within the allocated storage, either an `int[3]` object or an `int[4]` object is created. Both are correct interpretations of the program, and naturally both result in the same behavior. We can choose to view the program as being in the superposition of those two states. If we add a fourth `push_back` call to the program prior to the initialization of `n`, then only the `int[4]` interpretation remains valid.

```
unique_ptr<char[]> Stream::read() {
    // ... determine data size ...
    unique_ptr<char[]> buffer(new char[N]);
    // ... copy data into buffer ...
    return buffer;
}

void process(Stream *stream) {
    unique_ptr<char[]> buffer = stream->read();
    if (buffer[0] == F00)
        process_foo(reinterpret_cast<Foo*>(buffer.get())); // #1
    else
        process_bar(reinterpret_cast<Bar*>(buffer.get())); // #2
}
```

Note the `new char[N]` implicitly creates objects within the allocated array. In this case, the program would have defined behavior if an object of type `Foo` or `Bar` (as appropriate for the content of the incoming data) were implicitly created *prior* to `Stream::read` populating its buffer. Therefore, regardless of which arm of the `if` is taken, there is a set of implicit objects sufficient to give the program defined behavior, and thus the behavior of the program is defined.

§ 3.8. Direct object creation

In some cases it is desirable to change the dynamic type of existing storage while maintaining the object representation. If the destination type is a trivially-copyable implicit-lifetime type, this can be accomplished by copying the storage elsewhere, using placement new of an array of byte-like type, and copying the storage back to its original location, then using `std::launder` to acquire a pointer to the newly-created object. However, for expressivity and optimizability, a combined operation to create an object of implicit-lifetime type in-place while preserving the object representation may be useful. For this we propose:


```
// Effects: create an object of implicit-lifetime type T in the storage
//          pointed to by T, while preserving the object representation.
template<typename T> T *start_lifetime_as(void *p);
```

Note that such an operation is not sufficient to implement `node_handle` ([P0083R3]) for map-like containers. `node_handle` requires the ability to take a `std::pair<const Key, Value>` and permit mutation of the `Key` portion (without destroying and recreating the `Key` object), even when `Key` is not an implicit-lifetime type, so the above operation does not quite suffice. However, we could imagine extending its semantics to also permit conversions where each subobject of non-implicit-lifetime type in the destination corresponds to an object of the same type (ignoring cv-qualifications) in the source.

§ 4. Disposition and shipping vehicle

This paper did not complete LWG review in time for C++20. However, the functionality contained herein can be split into two portions:

- The core language change that gives defined behavior to various constructs that have historically been assumed to work, and
- The standard library addition of `std::start_lifetime_as`, which is a pure extension.

The author suggests that the committee considers adopting the former portion of this paper as a Defect Report, for possible inclusion into the C++20 IS, and that the latter portion be deferred to C++23.

Wording is presented assuming the committee agrees with this direction.

The portions of the wording below corresponding to the library extension are presented in this style, and marked "**Library extension, not for inclusion in C++20**".

Such wording is presented only for context and should not be applied to the C++20 working draft.

§ 5. Wording

§ 5.1. 6.6.2 Object model [intro.object]

Change in 6.6.2 [intro.object] paragraph 1:

The constructs in a C++ program create, destroy, refer to, access, and manipulate objects. An *object* is created by a definition (6.1), by a new-expression (7.6.2.4), by an operation that implicitly creates objects (see below), when implicitly changing the active member of a union (10.4), or when a temporary object is created (7.3.4, 6.6.7). [...]

Add a new paragraph at the end of [intro.object]:

Some operations are described as *implicitly creating objects* within a specified region of storage. For each operation that is specified as *implicitly creating objects*, that operation implicitly creates and starts the lifetime of zero or more objects of implicit-lifetime types (6.7 [basic.types]) in its specified region of storage if doing so would result in the program having defined behavior. If no such sets of objects would give the program defined behavior, the behavior of the program is undefined. [Note: Such operations do not start the lifetimes of subobjects of such objects that are not themselves of implicit-lifetime types. -end note]

Add an example following the new paragraph:

[Example:

```
#include
struct X { int a, b; };
X *make_x() {
    // The call to std::malloc implicitly creates an object of type X
    // and its subobjects a and b in order to give the subsequent
    // class member access operations defined behavior.
    X *p = (X*)std::malloc(sizeof(struct X));
    p->a = 1;
    p->b = 2;
    return p;
}
```

-- end example]

Add another paragraph:

An operation that begins the lifetime of an array of `char`, `unsigned char`, or `std::byte` implicitly creates objects within the region of storage occupied by the array. [Note: the array object provides storage for these objects. -- end note] Any implicit or explicit invocation of a function named `operator new` or `operator new[]` implicitly creates objects in the returned region of storage. Some functions in the C++ standard library implicitly create objects (19.10.9.2 [allocator.traits.members], 19.10.12 [c.malloc], 20.5.3 [cstring.syn], 26.5.3 [bit.cast]).

Library extension, not for inclusion in C++20.

Change in the newly-added paragraph:

Some functions in the C++ standard library implicitly create objects (16.6.x [obj.lifetime], 19.10.9.2 [allocator.traits.members], 19.10.12 [c.malloc], 20.5.3 [cstring.syn], 26.5.3 [bit.cast]).

§ 5.2. 6.6.3 Object and reference lifetime [basic.life]

Change in 6.6.3 [basic.life] paragraph 1:

[...] The lifetime of an object of type `T` begins when: [...] except that if the object is a union member or subobject thereof, its lifetime only begins if that union member is the initialized member in the union (9.3.1, 11.9.2), or as described in 11.4 and [class.copy.ctor] . [...] The lifetime of an object `o` of type `T` ends when:

- if `T` is a non-class type, the object is destroyed, or
- if `T` is a class type, the destructor call starts, or
- the storage which the object occupies is released, or is reused by an object that is not nested within `o` (6.6.2).

§ 5.3. 6.7 Types [basic.types]

Change in 6.7 [basic.types] paragraph 9:

[...] Scalar types, trivial class types (10.1), arrays of such types and cv-qualified versions of these types are collectively called *trivial types*. Scalar types, standard-layout class types (10.1), arrays of such types and cv-qualified versions of these types are collectively called *standard-layout types*. Scalar types, implicit-lifetime class types ([class.prop1]), array types, and cv-qualified versions of these types are collectively called *implicit-lifetime types*.

§ 5.4. 7.5.4.3 Destruction [expr.prim.id.dtor]

Change in 7.5.4.3 [expr.prim.id.dtor] paragraph 2:

If the *id-expression* names a pseudo-destructor, T shall be a scalar type and the *id-expression* shall appear as the right operand of a class member access (7.6.1.4) that forms the *postfix-expression* of a function call (7.6.1.2). [Note: Such a call ~~has no effect~~ ends the lifetime of the object ([expr.call], [basic.life]). —end note]

§ 5.5. 7.6.1.2 Function call [expr.call]

Change in 7.6.1.2 [expr.call] paragraph 5:

[...] If the *postfix-expression* names a *pseudo-destructor*, the *postfix-expression* must be a possibly-parenthesized class member access (7.6.1.4), and the function call ~~has no effect~~ destroys the object of scalar type denoted by the object expression of the class member access.

§ 5.6. 10.1 Properties of classes [class.prop]

Add a new paragraph at the end of [class.prop]:

A class S is an *implicit-lifetime class* if it is an aggregate ([dcl.aggr]) or has at least one trivial eligible constructor and a trivial, non-deleted destructor.

§ 5.7. 11.3.4.2 Copy/move constructors [class.copy.ctor]

Change in 11.3.4.2 [class.copy.ctor] paragraph 15:

The implicitly-defined copy/move constructor for a union X copies the object representation (6.7) of x .
For each object nested within ([intro.object]) the object that is the source of the copy, a corresponding object o nested within the destination is identified (if the object is a subobject) or created (otherwise), and the lifetime of o begins before the copy is performed.

§ 5.8. 11.3.5 Copy/move assignment operator [class.copy.assign]

Change in 11.3.5 [class.copy.assign] paragraph 13:

The implicitly-defined copy assignment operator for a union X copies the object representation (6.7) of x .

If the source and destination of the assignment are not the same object, then for each object nested within ([intro.object]) the object that is the source of the copy, a corresponding object o nested within the destination is created, and the lifetime of o begins before the copy is performed.

§ 5.9. 19.10.9.2 Static member functions [allocator.traits.members]

Add paragraph after 19.10.9.2 [allocator.traits.members] paragraph 1:

Remarks: Implicitly creates objects (6.6.2 [intro.object]) in the returned region of storage.

Add paragraph after 19.10.9.2 [allocator.traits.members] paragraph 2:

Remarks: Implicitly creates objects (6.6.2 [intro.object]) in the returned region of storage.

5.10. 19.10.12 C library memory allocation [c.malloc]

Add a new paragraph after [c.malloc] paragraph 4 in the description of `aligned_alloc`, `calloc`, `malloc`, and `realloc`:

These functions implicitly create objects (6.6.2 [intro.object]) in the returned region of storage. In the case of `calloc` and `realloc`, the objects are created before the storage is zeroed or copied, respectively.

5.11. 20.5.3 Header `<cstring>` synopsis [cstring.syn]

Change in 20.5.3 [cstring.syn] paragraph 3:

The functions `memcpy` and `memmove` are signal-safe (16.12.4). Both functions implicitly create objects (6.6.2 [intro.objects]) in the destination region of storage immediately prior to copying the sequence of characters to the destination.

Library extension, not for inclusion in C++20.

5.12. 20.10.2 Header `<memory>` synopsis [memory.syn]

Add the following after the declarations of `std::align` and `std::assume_aligned`:

```
// [obj.lifetime] Explicit lifetime management
template T *std::start_lifetime_as(void *p);
template volatile T *std::start_lifetime_as(volatile void *p);
```

Add the following subclause immediately after 20.10.6 [ptr.align]:

5.13. 20.10.x Explicit lifetime management [obj.lifetime]

```
template T *std::start_lifetime_as(void *p);
template volatile T *std::start_lifetime_as(volatile void *p);
```

Mandates: T is an implicit-lifetime type.

Requires: `[p, (char*)p + sizeof(T))` denotes a region of allocated storage that is a subset of the region of storage reachable through p.

Effects: Implicitly creates objects within the denoted region, including an object A of type T whose address is p. The object representation of A is the contents of the storage prior to the call to `start_lifetime_as` as if by calling `std::memcpy` from a copy of the original storage, except that the storage is not accessed.

Returns: A pointer to A.

5.14. 26.5.3 Function template `bit_cast` [bit.cast]

Change in 26.5.3 [bit.cast] paragraph 1:

```
template<class To, class From>
constexpr To bit_cast(const From& from) noexcept;
```

Returns: An object of type To.

Implicitly creates objects in the result (6.6.2 [intro.object]). Each bit of the value representation of the result is equal to the corresponding bit in the object representation of `from`. Padding bits of the To object are unspecified. For the result and each object created within it, if `if` there is no value of the object's type `Fe` corresponding to the value representation produced, the behavior is undefined. If there are multiple such values, which value is produced is unspecified.

§ 5.15. C.5 C++ and ISO C++ 2017 [diff.cpp17]

Add an entry to Annex C as follows:

Affected subclause: [basic.life]

Change: A pseudo-destructor call ends the lifetime of the object to which it is applied.

Rationale: Increase consistency of the language model.

Effect on original feature: Valid ISO C++ 2017 code may be ill-formed or have undefined behavior in this International Standard.

```
constexpr int f() {
    int a = 123;
    using T = int;
    a.~T();
    return a; // undefined behavior; previously returned 123
}
static_assert(f() == 123); // ill-formed; previously valid
```

§ 5.16. Feature test macro

No feature test macro is proposed for the core language changes.

Library extension, not for inclusion in C++20.

For the library functionality, add feature test macro `__cpp_lib_start_lifetime_at` for header `<memory>` with a suitable value to Table 36 in 17.3.1 [support.limits.general].

§ 6. Acknowledgments

Thanks to Ville Voutilainen for raising this problem, and to the members of SG12 for discussing possible solutions.

§ References

§ Informative References

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[P0593R3]

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